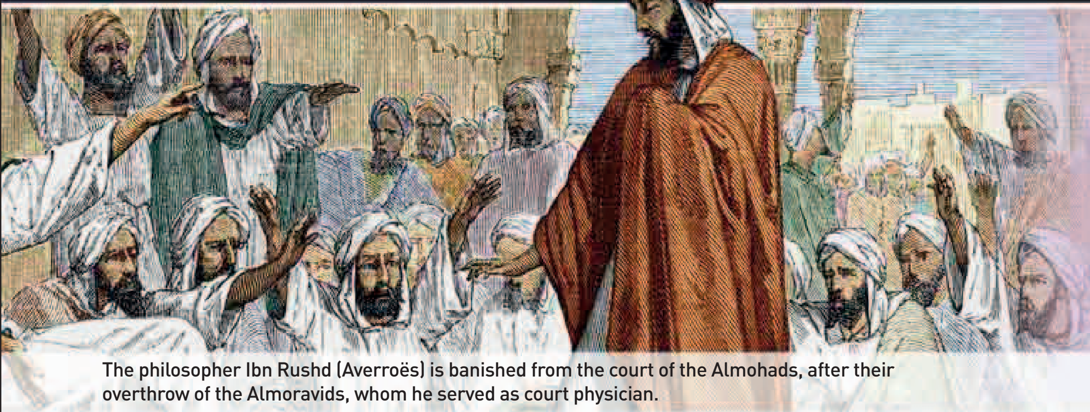
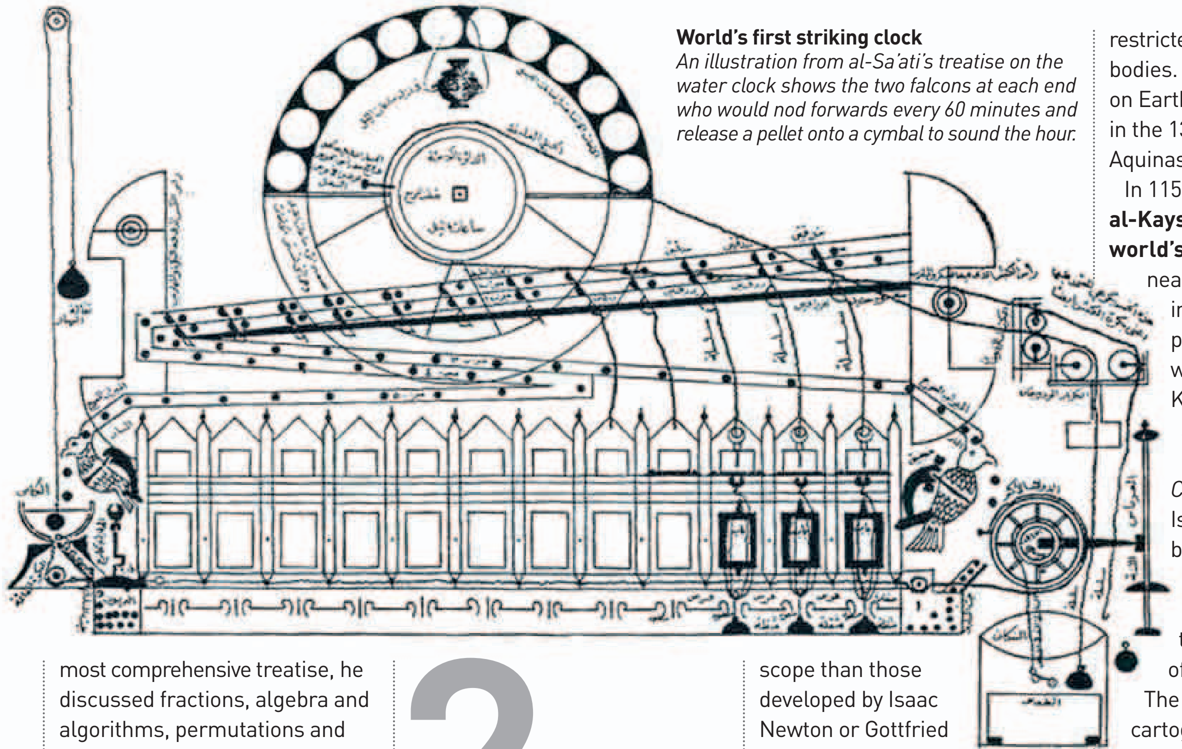


“ KNOWLEDGE IS THE CONFORMITY OF THE **OBJECT** AND THE **INTELLECT.** ”

Ibn Rushd (Averroës), from *Commentaries on the Physics*, late 12th century

The philosopher Ibn Rushd (Averroës) is banished from the court of the Almohads, after their overthrow of the Almoravids, whom he served as court physician.



World’s first striking clock
An illustration from al-Sa’ati’s treatise on the water clock shows the two falcons at each end who would nod forwards every 60 minutes and release a pellet onto a cymbal to sound the hour.

most comprehensive treatise, he discussed fractions, algebra and algorithms, permutations and combinations, and the geometry of triangles and quadrilaterals. He also introduced the idea of negative quantities in geometry. In his *Bija-Ganita* (*Seed Counting*), he concluded that the **division of a number by zero would produce infinity**. He also became the first mathematician to realize that there are **two square roots of a number**, one positive and one negative. In his astronomical work of 1150, the *Siddhanta-siromani* (*Head Jewel of Accuracy*), Bhaskara II performed calculations on small increments

2
THE NUMBER OF **SQUARE ROOTS** OF ANY NUMBER

of motion that came close to **an idea of differential calculus**, which studies the rates at which quantities change. However, his ideas were of much narrower scope than those developed by Isaac Newton or Gottfried Leibniz five centuries later.

The Spanish-born philosopher **Ibn Rushd** (1126–98), known as Averroës in Europe, commented extensively on Aristotle’s work in the 4th century BCE, seeking to integrate his ideas with Islamic theology. Around 1154, in his work on Aristotle’s theory of motion, Averroës made a distinction for the first time between the **motive force of an object** (its weight) **and the inherent resistance of a body to motion** (its mass), although he restricted this analysis to celestial bodies. Its extension to bodies on Earth would be made only in the 13th century by Thomas Aquinas (*c.*1224–74).

In 1154, Arab engineer **al-Kaysarani** constructed **the world’s first striking clock**, near the Umayyad mosque in Damascus. It was powered by water and was described by al-Kaysarani’s son Ridwan al Sa’ati in his 1203 treatise *On the Construction of Clocks and their Use*. Islamic water clocks became so sophisticated that in 1235 one was built in Baghdad that told people the times of prayer, day and night.

The advanced state of both cartography and printing in China are indicated by **the first printed map**, which dates from around 1155 (at least three centuries before its first European counterpart, in 1475). Contained in the *Liu Ching Tu* (*Illustrations of Objects mentioned in the Six Classics*), it depicted parts of western China with **rivers and provincial names given**, and showed the line of the Great Wall.

A more grandiose cartographic creation of the Chinese Sung dynasty was the *Yu Ji Tu*, an 1137 **map of the country carved in stone**, which included grid lines and an indication of the scale of the map.

While waterwheels had long been used in Europe for the grinding of grain, around 1180 the idea was adapted to the use of **windpower**. Unlike earlier Persian windmills, which were horizontal, the **European vertical mills used a post design, with sails** mounted on a vertical tower that itself was free to rotate as the wind varied. By the 1190s, windmills had become so commonplace that Pope Celestine III imposed a tax on them.

Vane power
This German windmill shows the typical arrangement of four sails attached to a vertical post, but unlike earlier post-mills only the cap of the mill rotates to face the wind.

1151–58 Abbess Hildegard of Bingen writes treatises on **medicine and botany**

1154 Ibn Rushd’s (Averroës’s) **theory of kinetics** expresses the idea that objects have mass that resists motive force

1154 Al-Kaysarani builds world’s **first striking clock** in Damascus

1155 First printed map, of part of western China

c.1155 French theologian Thierry of Chartres writes works reconciling Biblical account of creation with Platonic cosmology

1167 Italian scholar Gerard of Cremona becomes head of Toledo translation school; **translates nearly 90 works** into Latin

c.1180 Horizontal **windmills** appear in northwestern Europe

c.1180 Italian scholar Burgundio of Pisa **translates works of Galen** into Latin

1191 Salerno is sacked by Emperor Henry VI, leading to **decline of medical school**

1198 Jewish physician Moses Maimonides writes treatise on **poisons and their antidotes**